

Aussie Primary Academy

Year 6 Science – Physical Sciences: Electrical Circuits

I can: explain how series and parallel circuits transfer and transform electrical energy!

Name: _____

Date: _____

Page 1 – Let's Explore!

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Big Idea

Electricity flows through a **circuit** — a complete loop from a power source, through wires, to a component (like a globe), and back. In a **series circuit** there is one loop; in a **parallel circuit** there are multiple loops, so components can work independently.

1 Vocabulary Bank

circuit

series

parallel

conductor

insulator

current

energy transfer

component

2 Circle: Conductor or Insulator?

- a) Copper wire Conductor / Insulator
- b) Rubber Conductor / Insulator
- c) Aluminium foil Conductor / Insulator
- d) Plastic Conductor / Insulator

3 Series or Parallel? Match the description to the circuit type

One path only — if one globe fails, all stop



Series

Multiple paths — each globe can work alone



Parallel

4 True or False

- a) Electricity needs a complete loop to flow. True / False
- b) In a series circuit, if one globe blows, the others still work. True / False
- c) Metals are usually good conductors of electricity. True / False

5 Draw and label a series circuit AND a parallel circuit with a battery, wires and 2 globes

Series



Parallel



6 Investigate: Predict and test

If one globe is removed from a series circuit, what happens to the other globe? What if it's removed from a parallel circuit?



7 Challenge Question

Why do most homes use parallel circuits instead of series circuits for their lights and power points?

Great Science Thinking!

High-Five Tips for Parents & Teachers

If you have simple circuit kits (batteries, wires, small globes) at home, building a real circuit together makes the concept click far faster than diagrams alone.

Use household examples — Christmas lights that all go out when one bulb blows are a great series circuit example your child has likely seen.

Ask 'why' after each answer, not just 'what' — understanding energy transfer matters more than memorising terms.

Point out conductors and insulators around the house (metal cutlery vs wooden spoon handles) during everyday tasks.

Remind your child that safety around real electricity always comes first — this worksheet is about the science, not real wiring.

✓ Answer Key

2. a) Conductor b) Insulator c) Conductor d) Insulator

3. One path only→Series, Multiple paths→Parallel

4. a) True b) False c) True

6. Series: other globe turns off too. Parallel: other globe stays on.

7. Parallel circuits let each light/appliance work independently, so one failure doesn't turn off the whole house